The Local Housing-Market Effects of Act 22/60 Investor Purchases: Data and Empirical Design

August 2026

1 Research question

When a tax-incentivized migrant (an Act 22/60 "Individual Investor" decree holder) buys a residential property in Puerto Rico, what happens to the housing market immediately around that property — to prices, and to who transacts?

2 Data construction

2.1 Identifying investor properties

- **Decree holders.** Government lists of individual grantees: 3,181 (Act 22) and 2,672 (Act 60) names; 5,770 distinct first–last pairs pooled.
- **CRIM cadastral registry.** Name searches against the CRIM parcel layer (via its ArcGIS REST service) matched 26,673 unique parcels, each with full attributes, centroid coordinates, and the parcel's *most recent* sale (price, date, buyer, seller). CRIM stores only the latest transaction per parcel.
- **Karibe property registry.** Registry name-search results; cadastral numbers extracted from registral descriptions and resolved in CRIM provide a second, largely non-overlapping source.
- **Match confidence.** Name→parcel matching is noisy for common names. A name returning *exactly one* property is treated as high-confidence ("unique match"): 810 parcels via CRIM, 755 via Karibe (surname-validated, geocoded), union **1,306 parcels** (259 cross-validated in both). The event studies use this base.
- **Coverage.** High-confidence identification covers 22.5% of searched names (26.7% including a validated multi-match recovery tier), versus 49–61% declared homeownership in the Act 22 annual reports. The gap is dominated by LLC/trust purchases (invisible to name search), name-format failures, and registry availability; the identified sample skews toward single-property, personal-name buyers.

2.2 Events

An **event** is a dated investor purchase: unique-match parcels with a valid CRIM sale date, collapsing same-day, same-location purchases (condo units bought together) into one event, giving 1,256 events (1,200 in the decree era, ≥ 2012, used in estimation). Purchases cluster heavily: the median event has 25 other events within 1 km (158 are fully isolated), so contamination is measured per event and used in robustness.

2.3 Outcomes: nearby sales

For each event, every CRIM parcel within 1,000 m with a recorded sale: 3.47M sale$\times$event pairs over 328K distinct parcels, each carrying distance, sale price/date/parties, property characteristics (lot size, assessed structure value, condo-unit and vacant-land indicators), and 2020 census tract. *Rings:* 0–250 m (treated), 250–400 m (buffer, dropped), 400–1,000 m (control). *Filters:* nominal transfers ($<$ \$10k), implausible dates, and sales of investor-matched parcels are excluded. *Caveat:* CRIM keeps only each parcel's last sale, so earlier sales are censored by later resales; the within-window, near-versus-far design mitigates this, and Figure 5 tests it.

2.4 Placebo events

1,500 purchases by non-corporate *individual* buyers in the investor price band (\$285K–\$3.9M, the p25–p95 of investor purchases), located within 250 m of a real event site (where the sales pool has full coverage) but more than three years apart in time. Far rings are truncated at 750 m for both the placebo run and a matched real-event baseline.

2.5 Buyer/seller origin proxy

Party names are classified against the Census 2010 surnames file (`pcthispanic`), taking the maximum across name tokens (robust to CRIM's inconsistent name ordering; appropriate under the Puerto Rican two-surname convention). Names with `pcthispanic` ≤ 20 are "non-Hispanic-named" (mainland proxy), ≥ 70 "Hispanic-named" (local proxy); corporate, ambiguous, and unmatched names are left unclassified, and the raw score is retained for re-thresholding. *Validity:* 67% of investor purchases have non-Hispanic-named buyers versus 10% of surrounding market sales. *Caveat:* the proxy captures *name* origin — stateside Puerto Ricans and other Hispanic-named mainlanders classify as local, so mainland penetration and displacement effects are understated.

2.6 Summary statistics

Table 1 summarizes the estimation samples: the ring-design sale sample, the Red Atlas tract price panel, the HMDA purchase panel, and the 2010 PRCS baseline tract characteristics used for balance and robustness.

Table 1: **Summary Statistics** This table presents summary statistics for the paper's main samples. Panel A: market sales within 1km of an identified decree-era investor purchase (ring design estimation sample: gap ring, junk/nominal prices, and investor-parcel sales excluded; events 2012+); name-classification indicators are defined over sales whose parties' surnames classify. Panel B: Red Atlas tract×year panel (transaction-count-weighted annual mean prices); identified purchases reported over the 235 treated tracts. Panel C: HMDA tract×year panel of originated first-lien owner-occupied 1–4 family purchases, 2012–2024. Panel D: 2010 PRCS tract characteristics (2020 boundaries, population-weighted).

	Mean (1)	Std. Dev. (2)	P25 (3)	Median (4)	P75 (5)	Observations (6)
Panel A: Ring-design sales (≤1km of an event)						
Sale price ($000s)	1,872.3	61,984.0	116.6	240.0	475.0	971,720
Log sale price	12.35	1.27	11.67	12.39	13.07	971,720
1{non-Hispanic-named buyer}	0.169	0.375	0.000	0.000	0.000	660,570
1{Hisp. seller → non-Hisp. buyer}	0.115	0.320	0.000	0.000	0.000	500,309
1{non-Hisp. seller → Hisp. buyer}	0.097	0.295	0.000	0.000	0.000	500,309
Panel B: Tract price panel (Red Atlas)						
Tract annual mean price ($000s)	–	–	–	–	–	0
Tract annual transactions	–	–	–	–	–	0
Identified purchases (treated tracts)	5.1	13.7	1.0	1.0	3.0	235
Panel C: HMDA purchases (2012–2024)						
Hispanic-borrower purchases	11.6	12.3	4.0	8.0	15.0	11,350
Non-Hispanic-borrower purchases	0.195	0.819	0.000	0.000	0.000	11,350
Mean borrower income ($000s)	54.7	79.8	34.8	43.9	58.0	11,100
Panel D: 2010 PRCS tract characteristics						
Median household income ($)	20,242	10,759	13,704	17,102	23,700	875
Median home value ($)	119,534	54,345	88,700	104,150	134,675	874
Median gross rent ($)	461	181	352	436	541	863
Poverty rate	0.457	0.165	0.347	0.473	0.576	877
BA+ share (25+)	0.208	0.128	0.120	0.175	0.257	878
Renter share	0.291	0.155	0.187	0.251	0.358	877
Vacancy rate	0.162	0.088	0.105	0.147	0.199	877
Seasonal-home share	0.032	0.060	0.006	0.017	0.036	877
Born-in-mainland share	0.050	0.025	0.031	0.047	0.064	878
Population	4,020	2,008	2,634	3,990	5,320	936

3 Empirical design

3.1 Panel construction

Because CRIM records one (the last) sale per parcel, sales are aggregated into an event×ring×period panel. Let e index events, $r \in \{\text{near}, \text{far}\}$ rings, and t calendar periods (years in the robustness suite). The cell outcome is the mean log price among the N_{ert} transactions in cell (e, r) in period t:

$$\bar{y}_{ert} \;=\; \frac{1}{N_{ert}} \sum_{j \in (e,r,t)} \ln P_j, \tag{1}$$

where P_j is the sale price of transaction j. Treatment is absorbing and turns on for the near cell in the event period t_e:

$$D_{ert} \;=\; \mathbf{1}\{r = \text{near}\} \times \mathbf{1}\{t \geq t_e\}, \tag{2}$$

so far cells are never treated and serve as clean controls. In the tract-fixed-effects specification, $\ln P_j$ is first residualized on census tract dummies at the sale level ($\ln P_j - \hat{\gamma}_{\tau(j)}$) and the residual is aggregated instead; results are unchanged.

3.2 Local projections difference-in-differences (LP-DiD)

Following Dube, Girardi, Jordà and Taylor (2023), the dynamic effect at each horizon h is estimated from a separate long-difference regression on the stacked cell panel. For post-treatment horizons $h = 0, 1, \ldots, H$:

$$\bar{y}_{er,\,t+h} - \bar{y}_{er,\,t-1} \;=\; \beta_h^{\text{LP-DiD}}\, \Delta D_{ert} \;+\; \delta_t^h \;+\; \varepsilon_{ert}^h, \tag{3}$$

and for pre-treatment horizons $h = 2, \ldots, Q$ (with $h = 1$ the normalization):

$$\bar{y}_{er,\,t-h} - \bar{y}_{er,\,t-1} \;=\; \beta_{-h}^{\text{LP-DiD}}\, \Delta D_{ert} \;+\; \delta_t^h \;+\; \varepsilon_{ert}^h, \tag{4}$$

where $\Delta D_{ert} = D_{ert} - D_{er,t-1}$ indicates that the near cell of event e enters treatment in period t, and δ_t^h are horizon-specific calendar-period effects.

Each horizon-h regression is estimated on the restricted "clean" sample

$$\Big\{ (e, r, t) : \quad \underbrace{\Delta D_{ert} = 1}_{\text{newly treated near cells}} \quad \text{or} \quad \underbrace{D_{er,s} = 0 \;\; \forall s}_{\text{never-treated far cells}} \quad \Big\}, \tag{5}$$

which is the `nevertreated` option: the counterfactual change $\bar{y}_{er,t+h} - \bar{y}_{er,t-1}$ for a newly treated near cell is identified from far cells over the same calendar window, and no already-treated observation is ever used as a control (the negative-weighting problem of two-way fixed effects under staggered timing cannot arise). $\beta_h^{\text{LP-DiD}}$ is a variance-weighted average of event-specific effects; standard errors are clustered by cell. The identifying assumption is within-micro-neighborhood parallel trends: absent the purchase, prices of homes transacting 0–250 m from the site would have evolved like those 400–1,000 m away, conditional on calendar-time effects — shocks common to the micro-area (Hurricane María, the 2020–21 in-migration wave) difference out by construction, and equations (4) provide the testable pre-trend restrictions $\beta_{-h} = 0$.

Two scope conditions follow from the ring structure. First, the near–far contrast differences out any effect *common to the full 1-km neighborhood*: the design identifies the hyper-local increment around a purchase site, not the neighborhood-level effect of aggregate investor inflows, which requires the complementary tract-level staggered design using the tract×month price panel (Section 5). Second, because purchases cluster at the enclave scale, other investor purchases in dense areas fall disproportionately inside the near ring itself; the estimated contrast there measures the cumulative local dose of investor activity rather than the effect of a single purchase (see Figure 5).

3.3 Robustness to the estimation approach

The cell-mean panel is a construction choice; the ring literature (Linden and Rockoff 2008; Campbell, Giglio and Pathak 2011) instead estimates on individual transactions: stacked sale-level regressions of log price on near×post (or near×event-time dummies) with event×ring and event×calendar-year fixed effects, standard errors clustered by event. The two implementations also weight differently — transactions versus events — so agreement is informative rather than mechanical. Re-estimating the baseline this way yields a pooled near×post effect of +7.09 log points (s.e. 1.58), and +6.72 (1.32) with hedonic controls (log lot size, sub-unit and vacant-land indicators), against the cell-level LP-DiD baseline of +6.95 (1.81); the event-study paths coincide horizon by horizon (Figure 1).

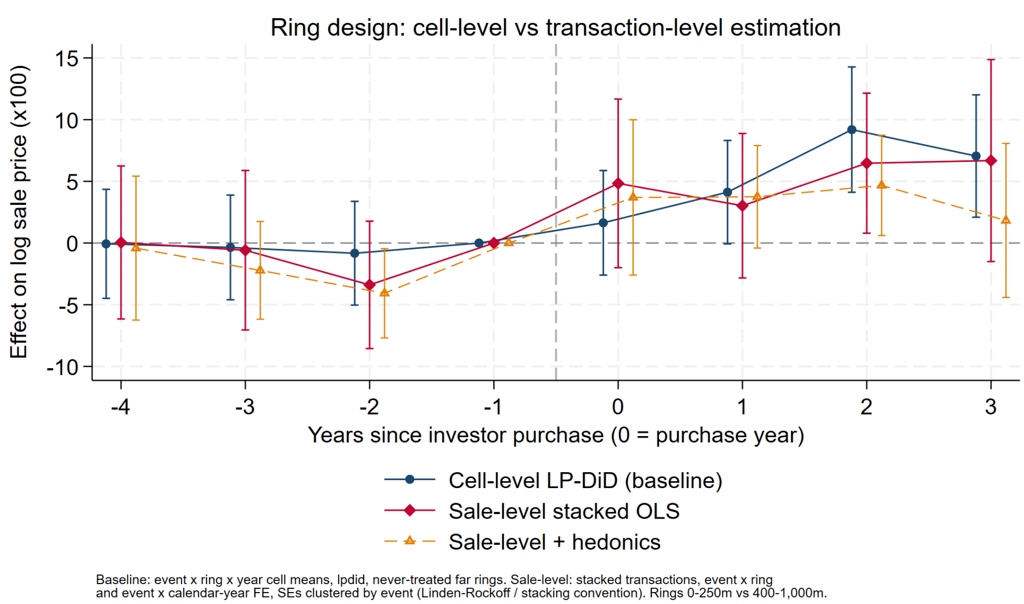

Figure 1: **Estimation-approach robustness.** The near-ring event study estimated three ways: the cell-level LP-DiD baseline; sale-level stacked OLS with event×ring and event×calendar-year fixed effects (the ring literature's convention), clustered by event; and the sale-level specification with hedonic controls. Rings 0–250 m vs 400–1,000 m. The three series coincide within sampling error at every horizon, and the agreement of transaction-weighted and event-weighted estimates indicates the pooled effect is not driven by a few high-volume events.

In the annual robustness suite $Q = 4$ and $H = 3$; the baseline is also estimated at quarterly ($Q = H = 8$) and half-yearly resolutions and with the Callaway–Sant'Anna (2021) group-time estimator $ATT(g,t) = \mathbb{E}[\bar{y}_t - \bar{y}_{g-1} \mid G_g = 1] - \mathbb{E}[\bar{y}_t - \bar{y}_{g-1} \mid \text{never treated}]$, aggregated to event time; results are similar across estimators and resolutions, and identical with and without tract fixed effects.

4 The eight figures

All figures plot LP-DiD event-study coefficients $\{\beta_{-Q}, \ldots, \beta_H\}$ from equations (3)–(4) at annual resolution (pre-window 4, post-window 3), with 95% confidence intervals, cells clustered. Period -1 is the omitted base; the dashed vertical line marks the treatment boundary.

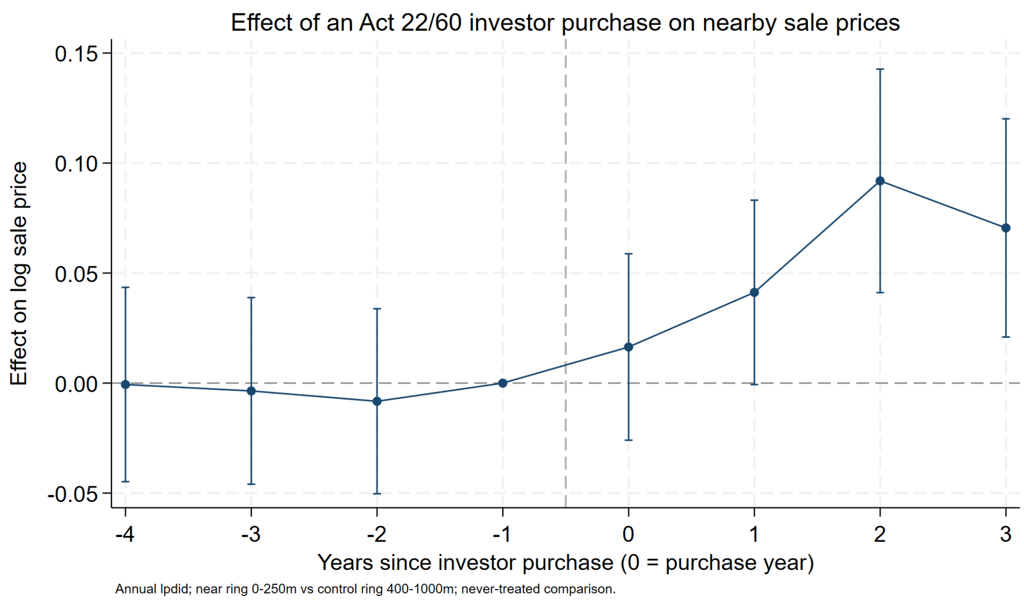

Figure 2: **Baseline event study.** Near-ring (0–250 m) versus far-ring (400–1,000 m) effects on log sale price by year relative to the investor purchase. The headline price bump; flat pre-period coefficients are the design's core credibility check, and the post-period path shows how quickly the premium appears and whether it persists.

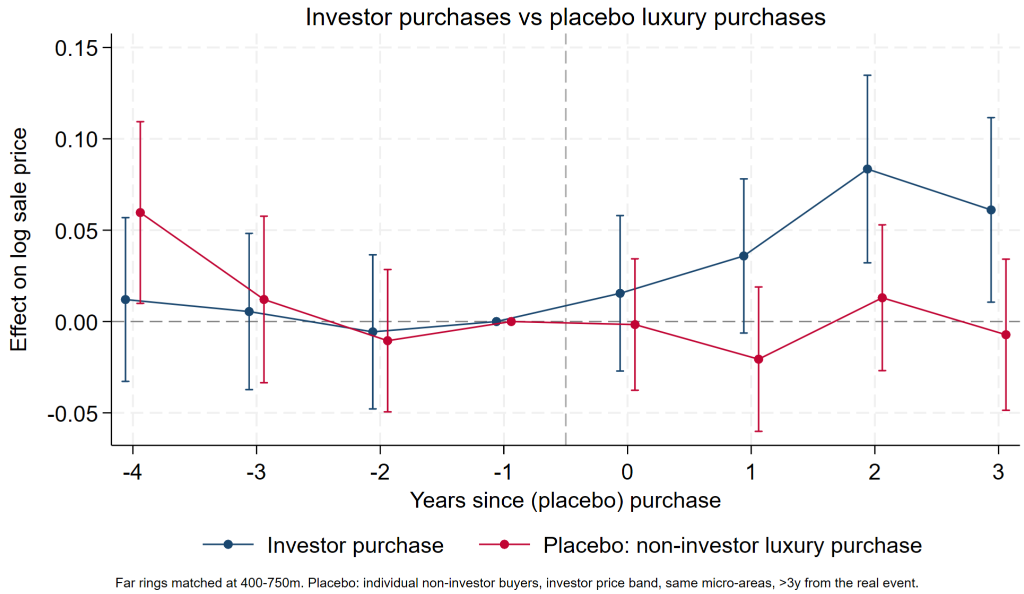

Figure 3: **Investor vs. placebo purchases.** The real event study overlaid with placebo events (non-investor individual buyers in the investor price band, same micro-areas, >3 years from the real event; far rings matched at 400–750 m). A flat placebo says nearby prices respond to *the investor*, not to any high-value transaction; because unmatched investors contaminate the placebo toward finding an effect, a null placebo is conservative evidence of investor specificity.

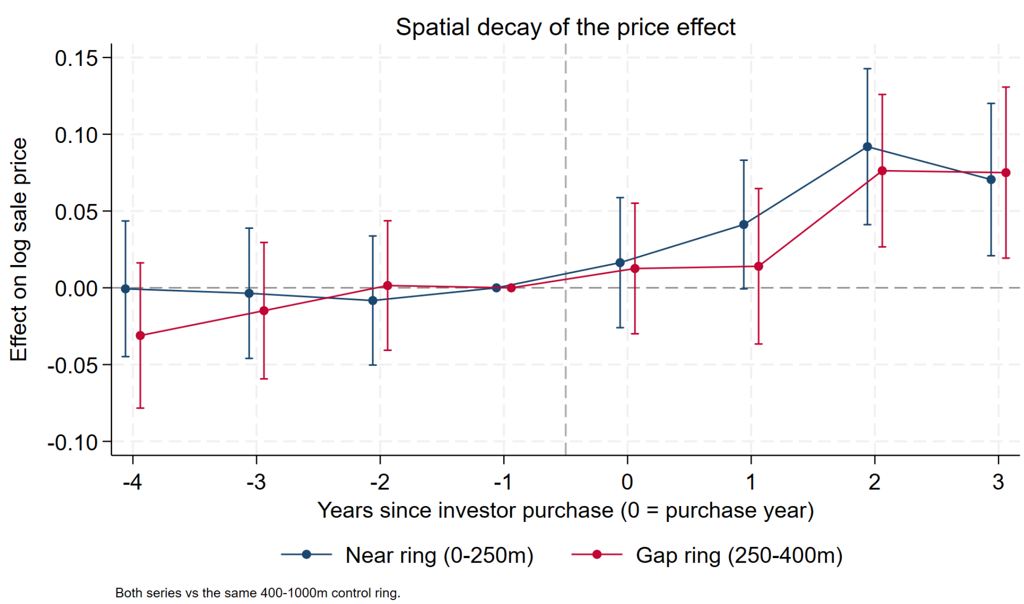

Figure 4: **Spatial gradient.** Near-ring (0–250 m) and gap-ring (250–400 m) series, each against the same far control. A genuine hyper-local spillover should decay with distance (near > gap > 0); near ≈ gap would instead indicate a neighborhood-wide shock partially shared by the control ring.

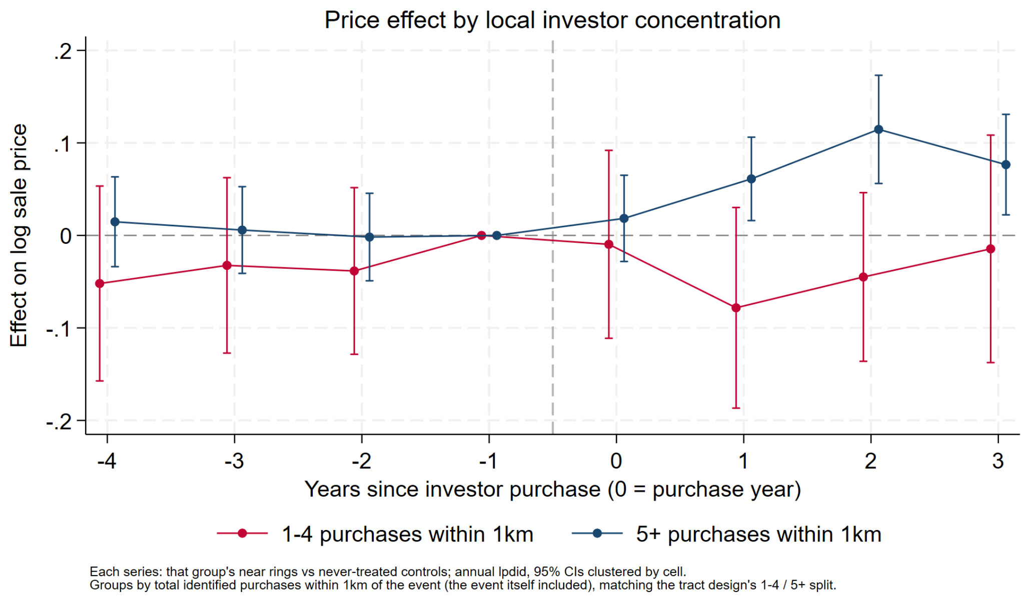

Figure 5: **Estimated contrast by local investor concentration.** Separate event studies for events with 1–4 vs. 5+ total identified purchases within 1 km (the event itself included), matching the tract design's dose split. Because purchases cluster at the enclave scale (condo buildings, resort communities), other events in dense areas fall disproportionately *inside* the near ring, so the near–far contrast there reflects cumulative local investor exposure rather than a single purchase. Low-concentration events show no price effect (post−pre −2.7, s.e. 3.5); 5+ areas show +6.4 (s.e. 1.5): single-purchase channels (comps entry, one renovation, signaling) appear weak on their own, and the price response is a phenomenon of investor *concentration* — the same 1–4/5+ pattern as the tract design. Pre-period coefficients are flat in both groups, inconsistent with investor waves selecting into already-appreciating pockets.

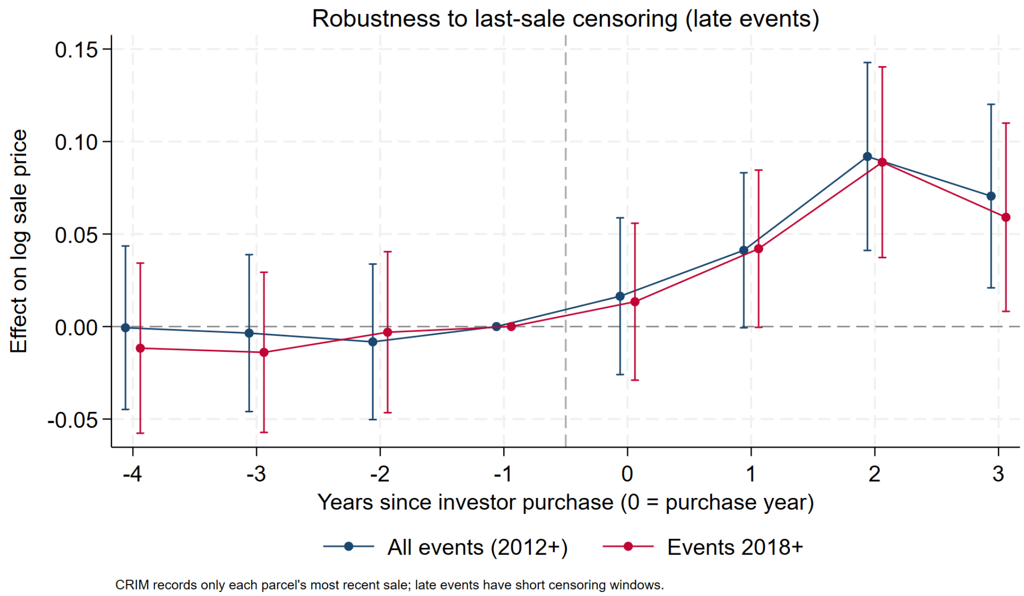

Figure 6: **Censoring robustness.** Baseline overlaid with events from 2018 onward, for which little time exists for post-period sales to be overwritten by later resales. Agreement with the full sample indicates that CRIM's last-sale-only censoring is not manufacturing the bump.

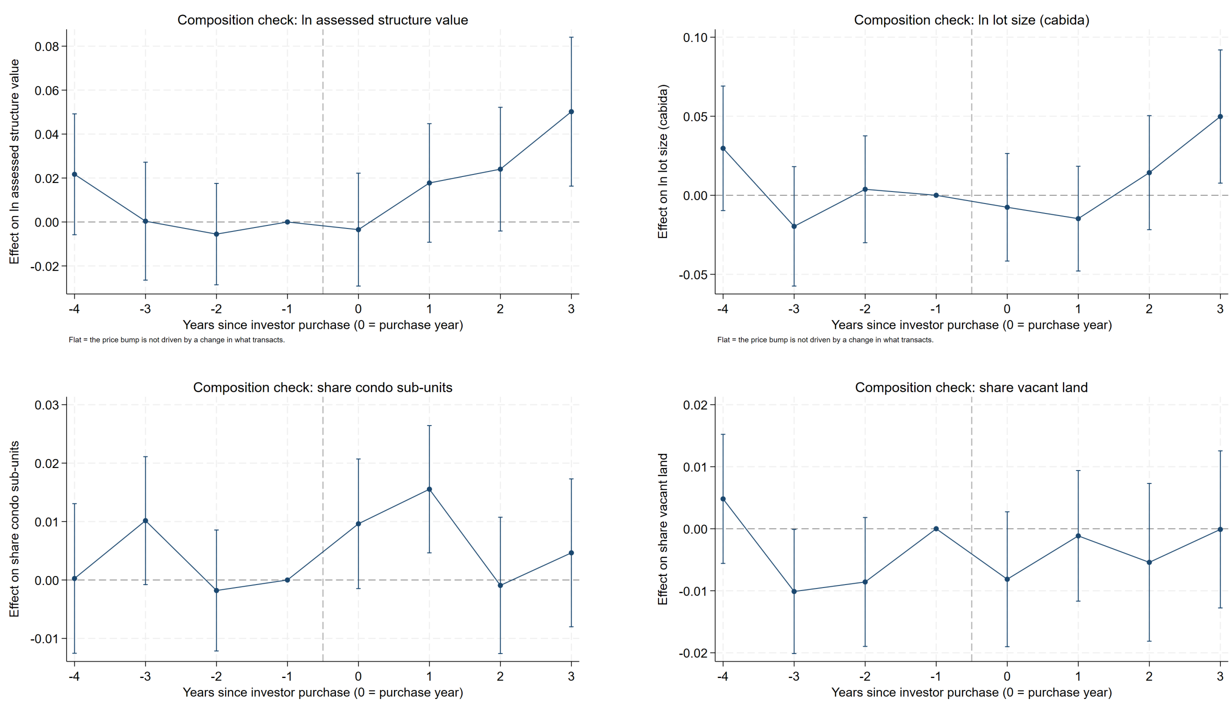

Figure 7: **Composition checks.** The identical design with property characteristics as outcomes: log assessed structure value (top left), log lot size (top right), condo-unit share (bottom left), vacant-land share (bottom right). Flat lines imply the price bump is appreciation on a stable transaction mix; movements would reveal the stock-upgrading margin of gentrification, changing the interpretation of Figure 2.

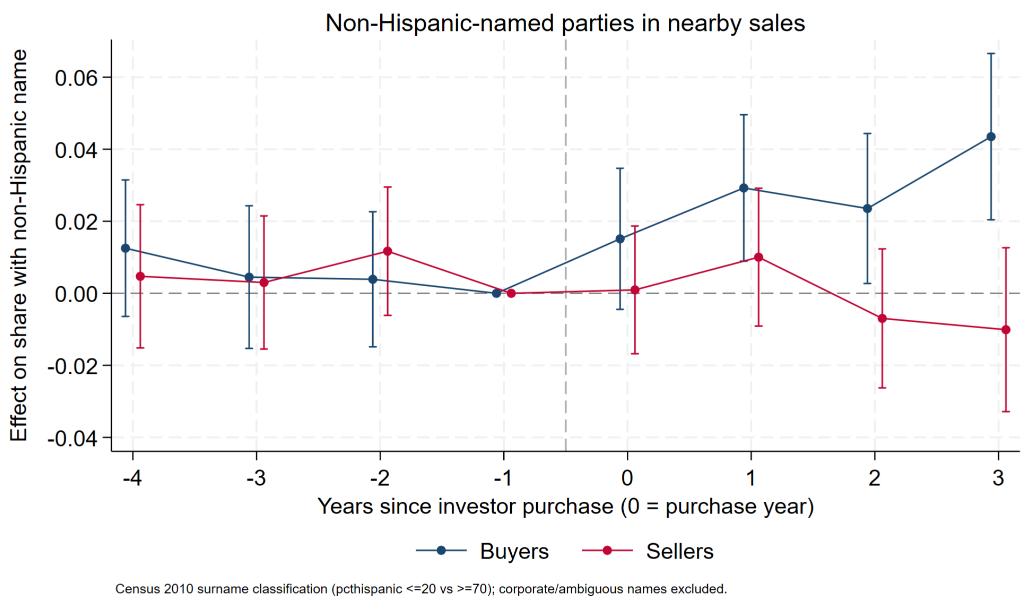

Figure 8: **Who transacts: buyers vs. sellers.** Effects on the share of nearby sales with non-Hispanic-named buyers and, separately, sellers (Census 2010 surname classification; corporate and ambiguous names excluded). Buyer share rising faster than seller share means the area is net-importing mainland owners after an investor arrives — the composition shift price indices cannot see.

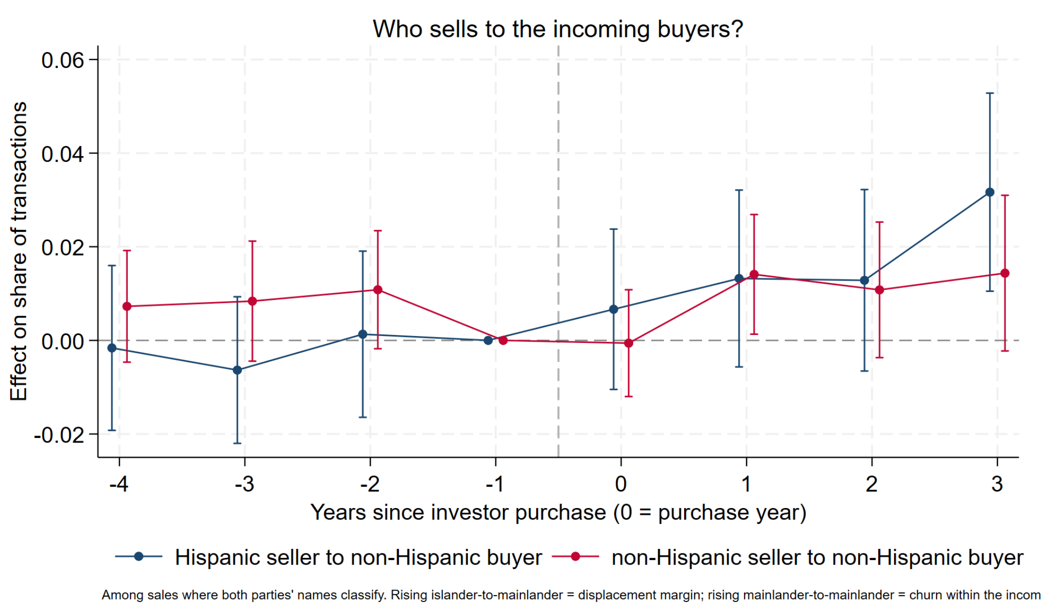

Figure 9: **Transition types.** Effects on the shares of transactions that are Hispanic-seller→non-Hispanic-buyer versus non-Hispanic-seller→non-Hispanic-buyer (among sales where both parties classify). Rising islander→mainlander transitions indicate conversion of locally held housing to incomers (the displacement margin); rising mainlander→mainlander indicates churn within an already-converted segment — together distinguishing "investors lead conversion" from "investors cluster where conversion already occurred."

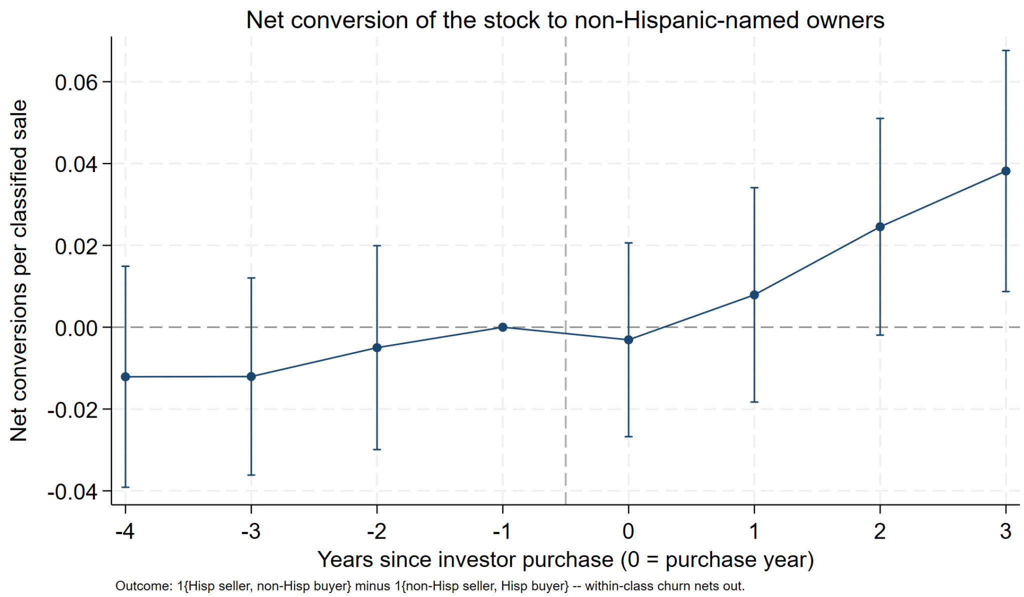

Figure 10: **Net compositional inflow.** The two transition series combined into one displacement measure: 1{Hispanic seller, non-Hispanic buyer} minus 1{non-Hispanic seller, Hispanic buyer} per classified sale. A sale changes the ownership composition of the stock only when the parties' classes differ, so within-class churn (including mainlander→mainlander resales) nets out; a positive post-event effect is net conversion of the nearby stock from Hispanic-named to non-Hispanic-named owners. Pre-period flat (-0.5pp, s.e. 1.0); the net conversion rate rises to $+2.5$pp and $+3.8$pp per sale in years 2–3; post$-$pre $+2.4$pp (s.e. 0.9, $p < 0.01$) per classified sale.

5 The tract-level design and reconciliation (Design 2)

5.1 Data and specification

The tract-level design asks what the ring design structurally cannot: whether investor purchases move *whole neighborhoods*, including the component common to treated areas that near–far contrasts difference out. The outcome panel is the Red Atlas tract×month transaction file (918 of 981 2020-vintage tracts, 2008–2025), aggregated to tract×year with the annual price defined as the transaction-count-weighted mean of monthly tract means (composition-noise reduction; the simple mean is a robustness variant). Treatment timing comes from the dated decree-era events: a tract is treated from the year of its first identified investor purchase (235 treated tracts; 1,200 events), with never-treated tracts as controls. Estimation is annual LP-DiD (pre-window 4, post-window 3) on $100 \times \ln(\text{price})$, with and without county fixed effects.

Table 2 compares treated and never-treated tracts on 2010 PRCS characteristics, all predetermined relative to Act 22. Treated tracts are richer (median income +\$5,700), higher-value (+\$39,000), more educated, less poor, and markedly more seasonal-home- and mainlander-oriented — exactly the profile of investor selection into coastal-amenity areas. Residualizing out municipio fixed effects shrinks the gaps only modestly (15–30%; all remain significant): treated tracts are distinct *within* municipios, not just across them. Selection on 2010 observables is therefore a fact to be handled, not assumed away — which is what Table 3 does, interacting the full 2010 covariate vector with the boom era on top of county×year fixed effects.

Table 2: **2010 Characteristics by Tract Treatment Status** This table presents summary statistics for 2010 PRCS (ACS 2006–2010 5-year) tract characteristics — all pre-determined relative to Act 22 (January 2012) — separated by tracts that ever receive an identified decree-era investor purchase (treated) and tracts that never do. Column (7) reports the raw mean difference between treated and never-treated tracts and column (8) the difference after residualizing out municipio (county) fixed effects, with heteroskedasticity-robust p-values in parentheses. Dollar variables in 2010 dollars; shares in [0,1]. 2010 tracts are mapped to 2020 boundaries by largest land overlap, population-weighted. Statistical significance levels are indicated as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

	Treated Tracts			Never-Treated Tracts			Mean Diff.	Adj. Mean Diff.
							[(1) – (4)]	[(1) – (4)]
	Mean	Std. Dev.	Observations	Mean	Std. Dev.	Observations	p-value	p-value
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Median household income ($)	24,522	13,955	218	18,821	9,039	657	+5,701***	+4,774***
							(0.000)	(0.000)
Median home value ($)	148,938	81,270	218	109,762	36,943	656	+39,176***	+28,126***
							(0.000)	(0.000)
Median gross rent ($)	515	206	216	442	169	647	+73***	+62***
							(0.000)	(0.000)
Poverty rate	0.400	0.169	218	0.476	0.160	659	-0.076***	-0.065***
							(0.000)	(0.000)
BA+ share (25+)	0.270	0.168	218	0.188	0.104	660	+0.082***	+0.071***
							(0.000)	(0.000)
Renter share	0.286	0.155	218	0.292	0.156	659	-0.006	-0.035***
							(0.594)	(0.003)
Vacancy rate	0.196	0.119	218	0.152	0.072	659	+0.044***	+0.037***
							(0.000)	(0.000)
Seasonal-home share	0.058	0.102	218	0.024	0.032	659	+0.034***	+0.030***
							(0.000)	(0.000)
Born-in-mainland share	0.057	0.032	218	0.047	0.022	660	+0.010***	+0.007***
							(0.000)	(0.002)
Population	4,549	1,935	218	3,859	2,004	718	+690***	+972***
							(0.000)	(0.000)

5.2 Results

Prices in treated tracts rise by a pooled +7.6 log points (s.e. 1.8) after the first investor purchase, with flat pre-period coefficients; adding county fixed effects leaves the estimate unchanged (+7.5), and the simple-mean construction gives +6.1 (Figure 11). Splitting treated tracts by investor concentration reproduces the dose pattern at tract scale: tracts with five or more identified purchases (44 tracts) show +18.3 (2.9), tracts with one to four show +3.8 (2.1) (Figure 12). Transaction *volume* falls in treated tracts, but its pre-period coefficients are significantly negative, so parallel trends fails for that outcome and we treat it as descriptive only.

Table 3 subjects the pooled tract-level estimates — prices, borrower income, and purchase counts by ethnicity — to a common specification ladder: (1) tract and year fixed effects, (2) tract and county×year fixed effects, and (3) additionally the ten standardized 2010 PRCS baseline characteristics interacted with the boom-era Post indicator. The price effect moves from +8.4 to

+7.2 to +5.6 (all $p < 0.01$), borrower income from +9.6 to +7.5 to +6.7, and non-Hispanic purchase entry from +37% to +26% to +25% ($p < 0.05$), while Hispanic purchase counts stay a precise null throughout. Panel C runs the same ladder on the ring design's event×ring cell panel (TWFE near×post coefficient; the pooled TWFE baseline of +5.1 is below the clean-control LP-DiD's +6.9 by construction): the price effect goes +5.1, +4.8, +3.1 (all $p < 0.05$) and the net ownership-conversion rate +1.8, +1.8, +1.6pp — the ring results, already identified off near–far contrasts, barely move when municipio-year shocks and baseline-trend selection are absorbed. Overall, roughly a quarter to a third of the raw estimates reflects county-level shocks and observable-trend selection; the remainder is identified within municipio-year among observably similar units.

Table 3: **Tract-level treatment effects are robust to county-year shocks and baseline-trend controls** This table reports pooled difference-in-differences estimates of the treatment indicator (1 from the tract's first identified decree-era purchase onward; never-treated tracts as controls) under three specifications per outcome: (1) tract and year fixed effects; (2) tract and county×year fixed effects; (3) adds ten standardized 2010 PRCS baseline characteristics (median household income, home value, and gross rent; poverty, BA+, renter, vacancy, seasonal-home, and mainland-born shares; log population) interacted with Post (1{year ≥ 2020}, the boom era). Panel A: tract log price (Red Atlas, count-weighted annual mean, 2012–2024) and log mean purchase-borrower income (HMDA, 2012–2024), both scaled by 100 (OLS). Panel B: purchase origination counts by borrower ethnicity (Poisson; coefficients ×100 ≈ percent effects, pseudo R-squared reported). Panel C: the ring design's event×ring cell panel (near ring × post-event indicator; cell fixed effects; the county and 2010 controls are the event's municipio and tract), for cell mean log price (×100) and the net Hispanic-to-non-Hispanic ownership conversion rate per classified sale (percentage points). Robust standard errors are clustered at the tract (Panels A–B) or cell (Panel C) level and reported in parentheses. Statistical significance levels are indicated as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Panel A: prices and borrower income (OLS)

	100 × Log(Price)			*100 × Log(Borrower Income)*		
	(1)	(2)	(3)	(1)	(2)	(3)
Treated	+8.45***	+7.20***	+5.59***	+9.56***	+7.54***	+6.69***
	(1.46)	(1.49)	(1.48)	(1.92)	(2.03)	(1.96)
Observations	16,343	16,324	15,314	11,088	11,071	10,707
R-squared	0.732	0.761	0.757	0.614	0.653	0.652
Tract FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	No	Yes	No	No
County-Year FE	No	Yes	Yes	No	Yes	Yes
2010 Controls × Post	No	No	Yes	No	No	Yes

Panel B: purchase originations by borrower ethnicity (Poisson)

	Hispanic Purchases			*Non-Hispanic Purchases*		
	(1)	(2)	(3)	(1)	(2)	(3)
Treated	+0.63	+1.67	+4.03	+37.38***	+26.06**	+24.59**
	(4.54)	(4.74)	(3.94)	(13.85)	(12.39)	(11.52)
Observations	12,350	12,329	11,159	7,033	5,379	4,991
Pseudo R-squared	0.549	0.565	0.548	0.353	0.375	0.383
Tract FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	No	Yes	No	No
County-Year FE	No	Yes	Yes	No	Yes	Yes
2010 Controls × Post	No	No	Yes	No	No	Yes

Panel C: ring design, event×ring cells (OLS)

	100 × Log(Price)			*Net H→NH Conversion (pp)*		
	(1)	(2)	(3)	(1)	(2)	(3)
Near × Post	+5.05***	+4.83***	+3.15**	+1.84**	+1.77**	+1.55**
	(1.36)	(1.23)	(1.29)	(0.75)	(0.72)	(0.74)

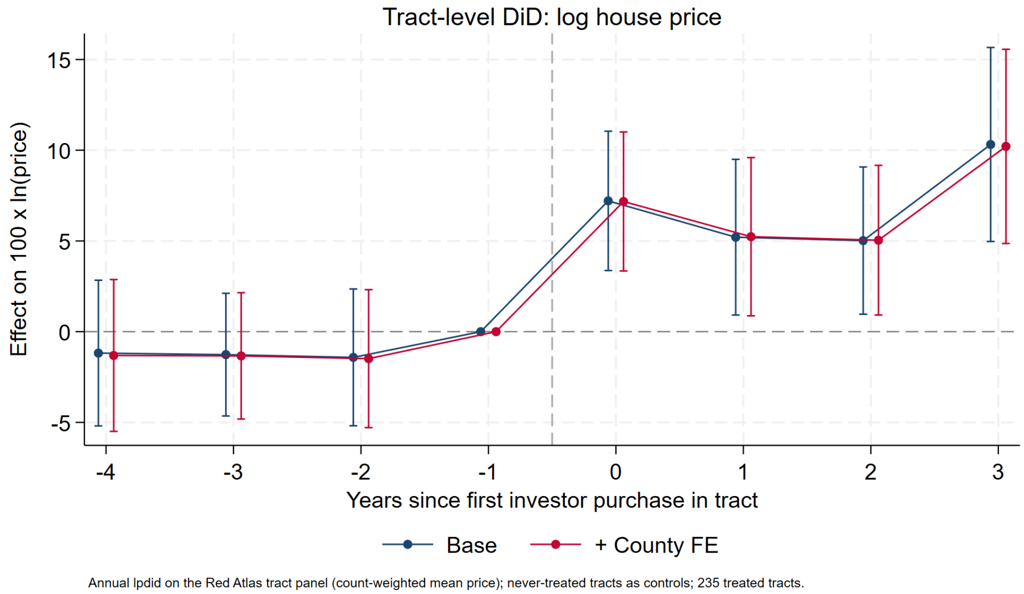

Figure 11: **Tract-level event study: log house price.** Annual LP-DiD on the Red Atlas panel, 235 treated tracts vs never-treated controls; base and county-FE series are indistinguishable and pre-trends are flat.

5.3 Reconciliation: the whole vs the sum of the parts

Because the ring design delivers a distance gradient, it implies a *prediction* for each treated tract: the stock-value-weighted average of $\beta(d)$ over the tract's parcels — the tract-level effect if micro-spillovers were the whole story. Comparing observed tract-DiD estimates to these predictions decomposes the aggregate (Figure 13): tracts with 1–4 purchases sit at or below their prediction (observed +3.8 vs predicted +5.4) — micro-spillovers fully account for them — while the 44 concentration tracts exceed their prediction by roughly 11 log points (observed +18.3 vs predicted +7.5). A market-level channel beyond proximity operates only where investors concentrate.

5.4 Aggregation

Integrating the estimated gradient over the exposure distribution of the island's improved housing stock (60.0% of parcels, 63.1% of assessed value, lie within 2.5 km of an identified event) yields an island-wide average price effect of +2.1 to +3.0%, or $6.3–8.0B of housing value at band-specific market/assessed ratios — roughly 6–8% of Puerto Rico's post-2019 boom (FHFA purchase-only index: +64% 2019–2024). Adding the concentration excess in the 44 high-dose tracts (≈$4.3B) raises the identified program footprint to ≈$11–12B, about 10–11% of the boom. At municipio level, using municipio repeat-sales indices as the local boom denominators, footprints in the top investor municipios are much larger (Table 4).

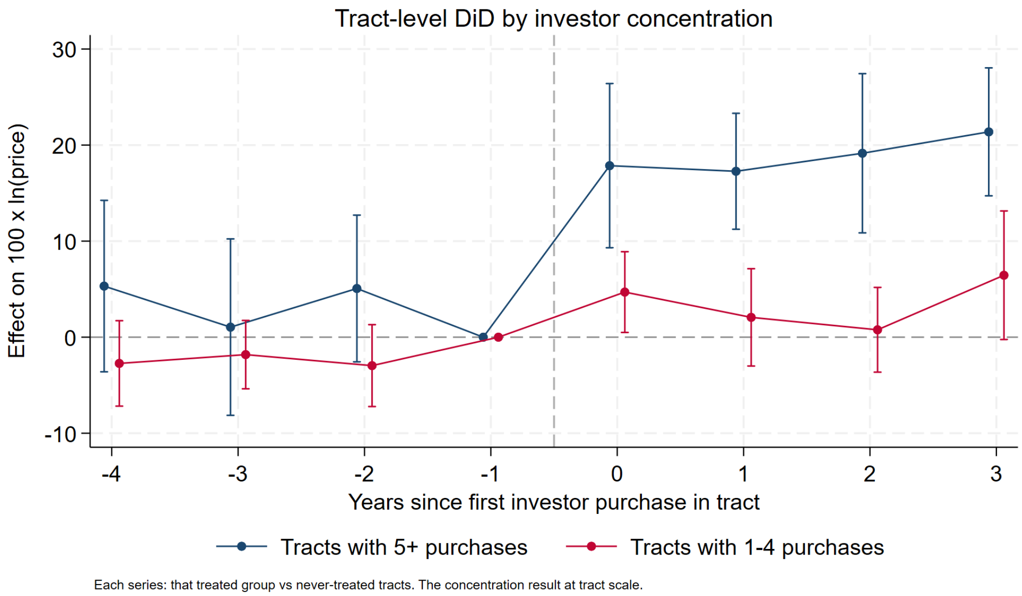

Figure 12: **Tract-level event study by investor concentration.** Tracts with 5+ identified purchases versus tracts with 1–4, each against never-treated controls: the concentration result at tract scale.

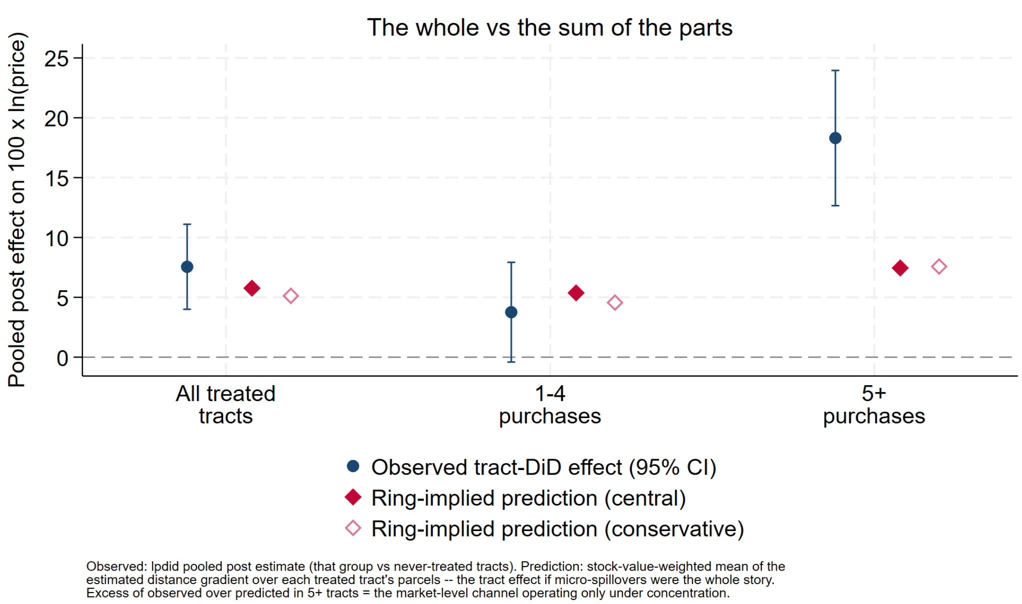

Figure 13: **Observed tract effects vs ring-implied predictions.** Observed: LP-DiD pooled post estimates by dose group. Predictions: stock-value-weighted distance-gradient effects over each treated tract's parcels. The all-treated and 1–4-purchase groups sit on their predictions; the 5+ group's excess (≈11 log points) is the concentration channel.

Table 4: **Program Footprint in the Most-Exposed Municipios** This table reports each municipio's post-2019 housing boom and the share of it attributable to identified investor purchases under three accountings. All columns share one valuation convention: each parcel at assessed value times its tract's median market/assessed ratio (2019+ sales; municipio and island fallbacks). The dollar boom is the municipio's market stock times $1 - 1/(1+g)$, the portion of current value attributable to repeat-sales growth g since 2019; Uplift is the synthesis in dollars. Shares of boom connect exactly: *Ring, Total* integrates the distance gradient over all parcels and *5+ tracts* restricts to parcels in 5+-purchase tracts; *Tract DiD* applies the dose-specific tract effects to treated-tract stock (*Total*: all treated tracts; *5+ tracts*: the high-dose subset). The synthesis *5+ differential* is Tract DiD (5+) minus Ring (5+) — the tract-wide repricing beyond what proximity predicts — and *Synthesis, Total* = Ring (Total) + the 5+ differential, so the ring layer is never double-counted.

	Local boom '19–'24			Synthesis		Ring design		Tract DiD	
Municipio	Growth	$	Uplift	Total	5+ diff.	Total	5+ tracts	Total	5+ tracts
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Rincón	+148%	$1.1B	$0.34B	31.1%	17.7%	13.4%	12.7%	30.7%	30.4%
Dorado	+177%	$3.7B	$0.97B	25.9%	14.4%	11.5%	10.3%	25.0%	24.8%
San Juan	+89%	$17.8B	$4.29B	24.1%	9.0%	15.1%	6.0%	17.8%	15.1%
Río Grande	+115%	$2.0B	$0.48B	24.0%	15.7%	8.4%	6.1%	23.5%	21.8%
Carolina	+91%	$7.6B	$1.32B	17.2%	6.8%	10.4%	4.7%	12.2%	11.6%
Guaynabo	+124%	$5.4B	$0.84B	15.5%	2.3%	13.2%	1.7%	8.6%	4.0%
Humacao	+121%	$2.5B	$0.39B	15.5%	7.2%	8.3%	5.1%	13.7%	12.4%

All aggregates are lower bounds: exposure counts only the 1,200 name-identified purchases, excluding LLC and unmatched investors; the capitalization step applies transaction-price effects to the full stock; and event effects predating 2019 enter the numerator while the boom clock starts in 2019.

5.5 Mortgage originations: volumes are a convincing null

HMDA loan-level data for Puerto Rico (2018–2024; 105,559 originations, aggregated to 2020-vintage tract×year) show no detectable response of mortgage lending in treated tracts: Poisson (PPML) estimates with tract and year fixed effects are statistically zero for owner-occupied purchases (−0.5%, s.e. 5.4), non-owner-occupied purchases (+3.0%, 11.3), rate/term and cash-out refinancings, and total originations, and are unchanged with county×year fixed effects. Consistent with decree investors transacting predominantly in cash, the program's effects appear in prices and buyer composition rather than local credit expansion.

5.6 Who can still buy: borrower composition in treated tracts

Splitting purchase originations by HMDA's self-reported borrower ethnicity (Hispanic or Latino vs. not; joint and unavailable excluded) sharpens the displacement interpretation. Since decree investors buy in cash, HMDA is effectively a census of the *financed* — overwhelmingly local, 94.6% Hispanic — side of the market. To get adequate pre-periods we extend the panel back to 2012 using

CFPB's historic HMDA files, on a population that is consistent across the 2017/2018 reporting change: originated first-lien, owner-occupied, 1–4 family home-purchase loans (2012–2024; annual volumes are smooth across the seam). Three results (Figures 14–15). First, borrower *income* rises after treatment — pooled +9.6% (s.e. 1.9), reaching +10.0% at $h \geq 3$ — and with up to six pre-years of data (displayed with the endpoint binned at −4) the pre-period is now flat (−3.7, −1.1, −2.4, individually and jointly insignificant, with no run-up into treatment; the apparent pre-drift in the short 2018–2024 panel was an artifact of its truncated window). Second, the income shift is concentrated where investors concentrate: +17.6% (s.e. 4.9) in 5+-purchase tracts vs. +7.9% (s.e. 2.0) in 1–4-purchase tracts. Third, on the extensive margin, Hispanic-borrower purchase counts are unchanged (pooled +0.6%, s.e. 4.5) while *non-Hispanic* borrower purchases rise +37% (s.e. 14) from a thin base — financed mainlander entry is now detectable, not just cash entry. Together: local buyers as a group keep transacting, but the marginal financed buyer moves up the income distribution, most strongly in high-dose tracts — the affordability-displacement margin, measured on self-reported ethnicity and income rather than the surname proxy.

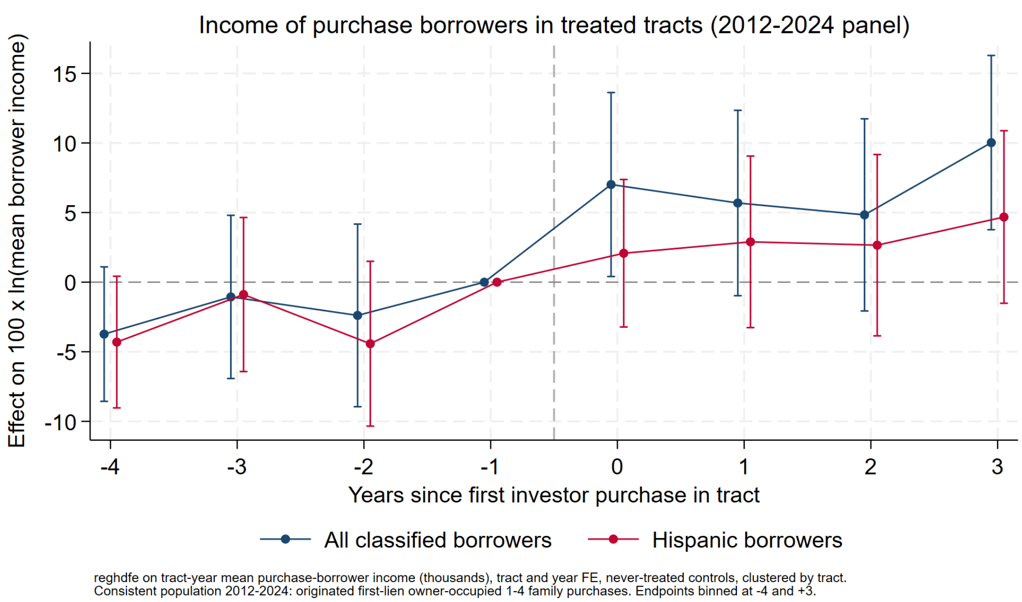

Figure 14: **Income of purchase borrowers, 2012–2024 panel.** Effect on $100 \times \ln$(mean borrower income) among first-lien owner-occupied purchase originations in treated tracts; all classified borrowers and Hispanic borrowers only. No pre-trend (endpoint binned at −4 pools horizons back to −6); pooled +9.6% (s.e. 1.9).

Financed mainlander entry

The non-Hispanic borrower series make the entry margin visible on its own terms (Figures 16–17; thin base — a few hundred non-Hispanic-borrower loans per year island-wide — hence the wide intervals). Purchase originations by non-Hispanic borrowers rise +37% (s.e. 14) pooled and +41% (s.e. 19) at $h \geq 3$ in treated tracts, with noisy but non-trending pre-periods: decree-era in-movers are

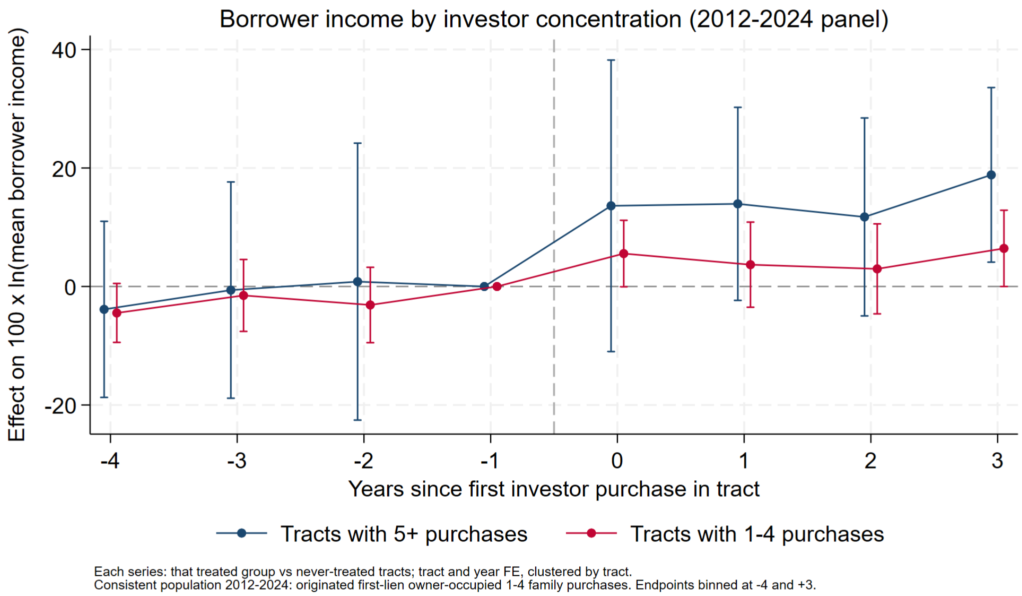

Figure 15: **Borrower income by investor concentration, 2012–2024 panel.** Each series: that treated group vs. never-treated tracts. The income shift is twice as large in 5+-purchase tracts (+17.6%, s.e. 4.9) as in 1–4-purchase tracts (+7.9%, s.e. 2.0) — the same concentration gradient as prices.

not exclusively cash buyers — a financed segment follows the same geography.[1] Their refinancings rise even more steeply — +70% (s.e. 17) pooled, +176% (s.e. 34) at $h \geq 3$ — consistent with newcomers refinancing recently acquired properties as the areas appreciate. (The corresponding *Hispanic* refinancing series is not usable as evidence: its pre-period climbs steadily into treatment with no post-treatment kink, so the incumbent equity-access margin is unidentified and we do not report it.)

6 Standing caveats

1. **Last-sale censoring** (CRIM): pre-period sales are survivors that never resold; addressed by design and Figure 5.

2. **Site selection**: investors choose appreciating micro-areas; estimates are local spillovers *conditional on* location choice, not "the effect of the Act."

[1] This magnitude reconciles with the ring design's buyer-composition effect once the two bases are compared. The CRIM non-Hispanic-*named* buyer share rises +2.5pp (s.e. 0.7, PMD post−pre) on a pre-treatment near-ring mean of 18.9% — nominally +13%. But the surname proxy inflates the base: self-reported non-Hispanic borrowers are only 1.6% of financed purchases in treated tracts pre-treatment, and non-Hispanic-*named* buyers are 10% of the general market against 67% among confirmed investors, so much of the 18.9pp proxy base is Hispanic-heritage locals with non-Hispanic-classifying surnames — a component inert to treatment. If the treatment-responsive (true mainlander) share of the proxy base is roughly a third of it (6–7pp), the +2.5pp effect is +35–40% of the responsive base, matching the HMDA estimate almost exactly. Both sources thus indicate mainlander purchasing rose by roughly one-third of its pre-treatment level — CRIM measuring all transactions at ring geography, HMDA the financed tail at tract geography.

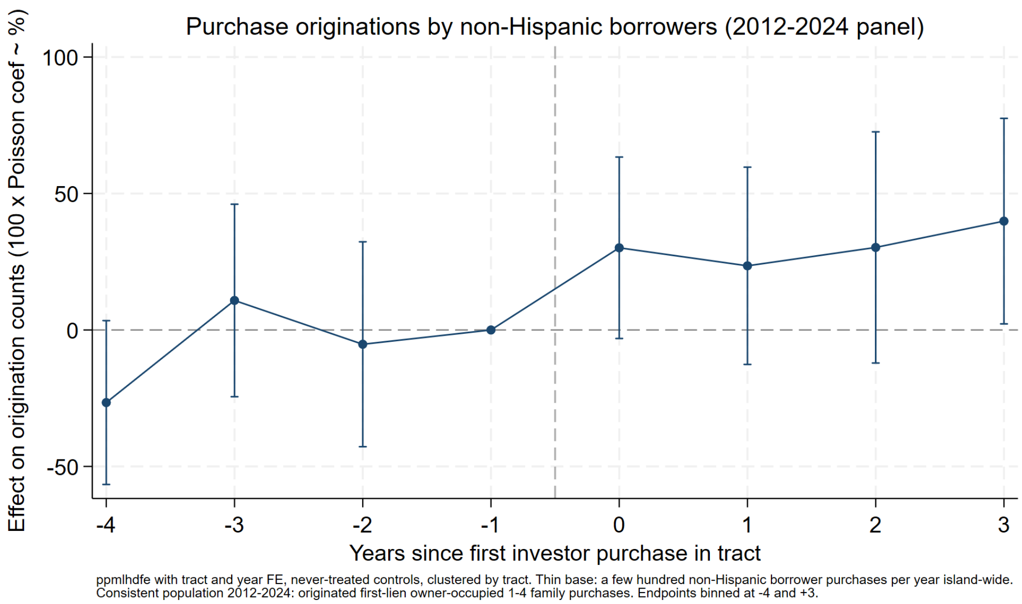

Figure 16: **Purchase originations by non-Hispanic borrowers, 2012–2024 panel.** Poisson event study, treated vs. never-treated tracts. Pooled +37% (s.e. 14).

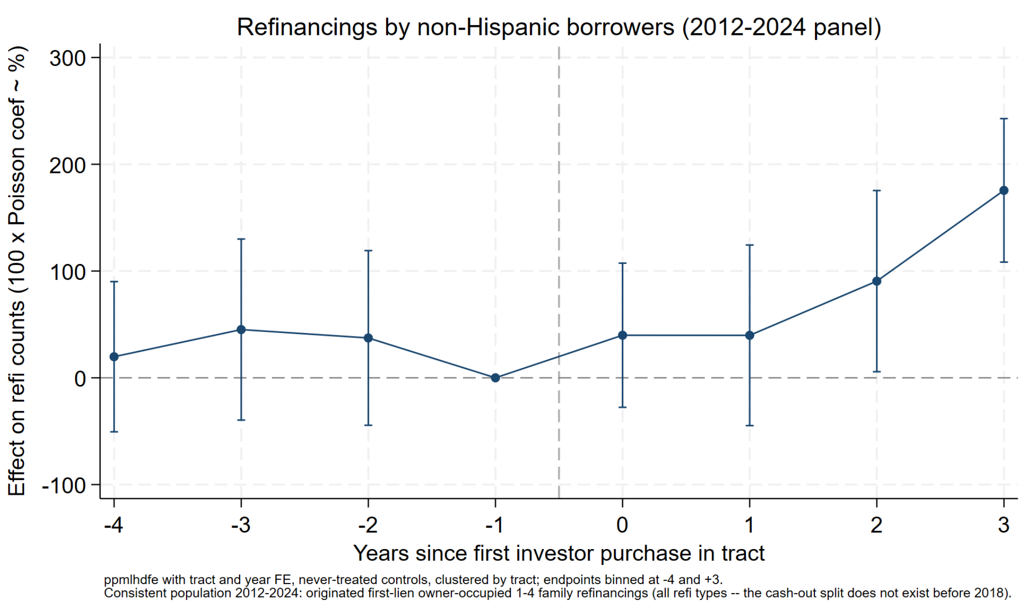

Figure 17: **Refinancings by non-Hispanic borrowers, 2012–2024 panel.** Same population (first-lien owner-occupied 1–4 family; all refi types — the cash-out split does not exist before 2018). Pooled +70% (s.e. 17), rising to +176% (s.e. 34) at $h \geq 3$.

3. **Match selection**: identified investors skew toward single-property, personal-name buyers; entity (LLC/trust) purchases are missing.
4. **Name proxy**: understates mainland presence; corporate and ambiguous names are excluded from shares.
5. **Assessed values** are on CRIM's 1957 basis — valid as cross-sectional controls and composition outcomes, never as market values.